

Goal Provision infrastructure on Verda Cloud and build a small Kubernetes platform that demonstrates practical DevOps/platform engineering skills. The assignment should take a few focused hours. We do not expect everything to be perfect. We care more about your reasoning, tradeoffs, debugging approach, and final summary than a copy-paste “happy path”.

Setup Signup and share you email and we will add credits on Verda Cloud account. Please provision CPU VMs with public IPs and build your environment there.

Scope:

1. Access instructions or screenshots
2. Git repository with manifests/scripts/docs
3. A short summary report:
 - What you built
 - Architecture diagram or explanation
 - What worked / what did not
 - Security and operational considerations
 - What you would improve with more time

Senior Assignment

Complete the base assignment, plus bonus task

1. Install Rancher Manager or explain clearly why you chose not to.
 - Configure any SSO if possible.
 - Register/import the created Kubernetes cluster.
2. Install Argo CD.
 - Configure SSO if possible.
 - Deploy at least one app through Argo CD.
 - Show GitOps structure and promotion thinking.
3. Add Harbor or another registry.
 - Push/build an image or explain image flow.
 - Connect deployment flow to Argo CD where possible.
4. Monitoring and operations.
 - Install kube-prometheus-stack or equivalent.
 - Add useful Grafana dashboards for cluster health.
 - Explain which alerts you would configure for production.

Optional Advanced Tasks Choose that fit the available time:

- Install Cilium and explain network policy / observability approach.
- Add KWOK to simulate a larger cluster, for example 100 nodes.
- Install Kueue and create high-priority and low-priority job queues.

- If GPU nodes are available:
 - Install NVIDIA drivers manually or with NVIDIA GPU Operator.
 - Trigger some GPU workload.
 - Show GPU scheduling and monitoring.
- Add backup strategy for Rancher, Argo CD, Harbor, and cluster state.
- Add security hardening